

```

1  package com.example;
2
3  public class Calculator {
4      public int add(int a, int b) {
5          return a + b;
6      }
7
8      public int max(int a, int b) {
9          if (a > b) {
10             return a;
11          } else {
12             return b;
13          }
14      }
15
16      public int divide(int a, int b) {
17          if (b == 0) {
18              throw new IllegalArgumentException(s: "Division by zero");
19          }
20          return a / b;
21      }
22  }
23

```

CalculatorTest.java ×

src > test > java > com > example > CalculatorTest.java > Language Support for Java(TM) by Red Hat > { } com.example

```

1  package com.example;
2
3  import static org.junit.jupiter.api.Assertions.assertEquals;
4  import static org.junit.jupiter.api.Assertions.assertThrows;
5  import org.junit.jupiter.api.Test;
6
7  public class CalculatorTest {
8      private final Calculator calculator = new Calculator();
9
10     @Test
11     public void testAddition() {
12         assertEquals(expected: 12, calculator.add(a: 5, b: 7));
13     }
14
15     @Test
16     public void testMaxTrueBranch() {
17         assertEquals(expected: 20, calculator.max(a: 20, b: 10));
18     }
19
20     @Test
21     public void testMaxFalseBranch() {
22         assertEquals(expected: 15, calculator.max(a: 5, b: 15));
23     }
24
25     @Test
26     public void testDivision() {
27         assertEquals(expected: 5, calculator.divide(a: 10, b: 2));
28     }
29
30     @Test
31     public void testDivisionByZero() {
32         Exception exception = assertThrows(InterruptedException.class, () -> calculator.divide(a: 10, b: 0));
33         assertEquals(expected: "Division by zero", exception.getMessage());
34     }
35 }

```

```
● PS C:\Users\Afroz\Desktop\calculator white boxtesting> mvn test
[INFO] Scanning for projects...
[INFO] -----< com.example:CalculatorProject >-----
[INFO] Building CalculatorProject 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- resources:3.4.0:resources (default-resources) @ CalculatorProject ---
[INFO] skip non existing resourceDirectory C:\Users\Afroz\Desktop\calculator white boxtesting\src\main\resources
[INFO] --- compiler:3.15.0:compile (default-compile) @ CalculatorProject ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- resources:3.4.0:testResources (default-testResources) @ CalculatorProject ---
[INFO] skip non existing resourceDirectory C:\Users\Afroz\Desktop\calculator white boxtesting\src\test\resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ CalculatorProject ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[INFO] --- surefire:3.2.5:test (default-test) @ CalculatorProject ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.CalculatorTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.049 s -- in com.example.CalculatorTest
[INFO] Results:
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.143 s
[INFO] Finished at: 2026-07-26T21:05:42+05:30
[INFO] -----
○ PS C:\Users\Afroz\Desktop\calculator white boxtesting> 
```